

DOES ADDITIVE NOISE DESTROY A PITCHFORK BIFURCATION?

LUCA MARTUCCI

CONTENTS

Introduction	1
1. Deterministic Dynamical Systems and their pitchfork bifurcations	4
2. Additive noise destroys a pitchfork bifurcation	7
3. Additive noise does NOT destroy a pitchfork bifurcation	13
Appendix	21
References	22

INTRODUCTION

It is well-known that certain classes of Ordinary Differential Equations, indexed by a parameter $\alpha \in \mathbb{R}$, display a pitchfork bifurcation for some critical value of the parameter. The canonical example, to which all pitchfork-bifurcating systems are topologically equivalent, is given by the class

$$\dot{x}_t = \alpha x_t - x_t^3 \quad \text{at } \alpha = 0$$

for trajectories $t \in [0, \infty) \mapsto x_t \in X = \mathbb{R}$. A possible physical motivation for such a dynamical system could be to study trajectories that move towards decreasing the potential

$$V(x) = -\frac{\alpha}{2}x^2 + \frac{1}{4}x^4$$

so that the equation takes the form of a gradient flow: $\dot{x}_t = -V'(x_t)$. These originate as an overdamped limit of Newton's equations, that describe the dynamics of a system under inertia and forces, governed by the Principle of Least Action: $m\ddot{x} = -\nabla V(x)$. The system can, in fact, be severely damped: $m\ddot{x} + \gamma\dot{x} = -\nabla V(x)$, for example by some friction or resistance, so that the inertial term becomes negligible and the equations of motion reduce to

$$\dot{x} = -\frac{1}{\gamma}\nabla V(x),$$

and we set $\gamma = 1$.

As the parameter varies and it crosses $\alpha = 0$, V goes from being a convex potential with a unique global minimum to a double-well potential, so that an energy barrier is created at the origin.

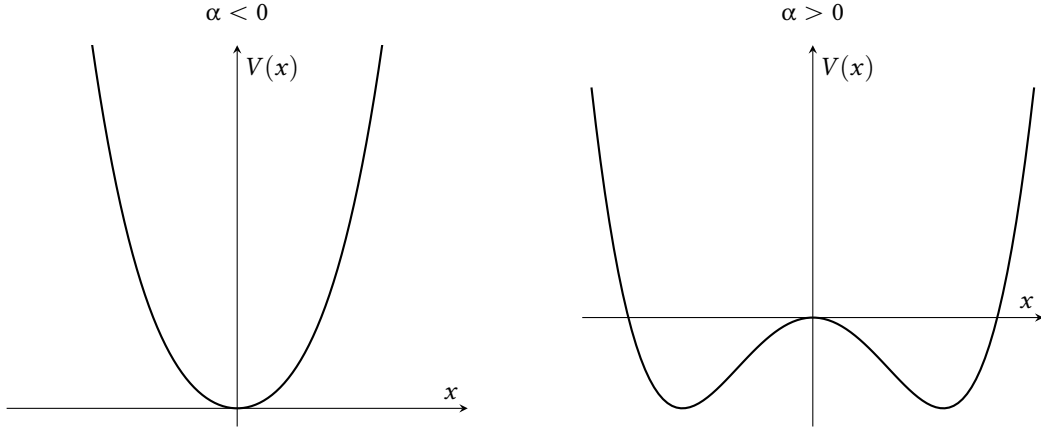


FIGURE 1. Potential landscape for a pitchfork bifurcation. Left: The origin is a unique stable minimum. Right: The origin becomes unstable, and two new stable minima appear.

At the level of the dynamics, minima correspond to stable equilibria of the system, as $\dot{x}_t = -V'(x_t) = 0$ and $\ddot{x}_t = -V''(x_t) < 0$, while maxima correspond to unstable fixed points. What we are observing in this didactic example is exactly the phenomenon of a pitchfork bifurcation: a pair of stable fixed points is created, at another fixed point that changes stability.

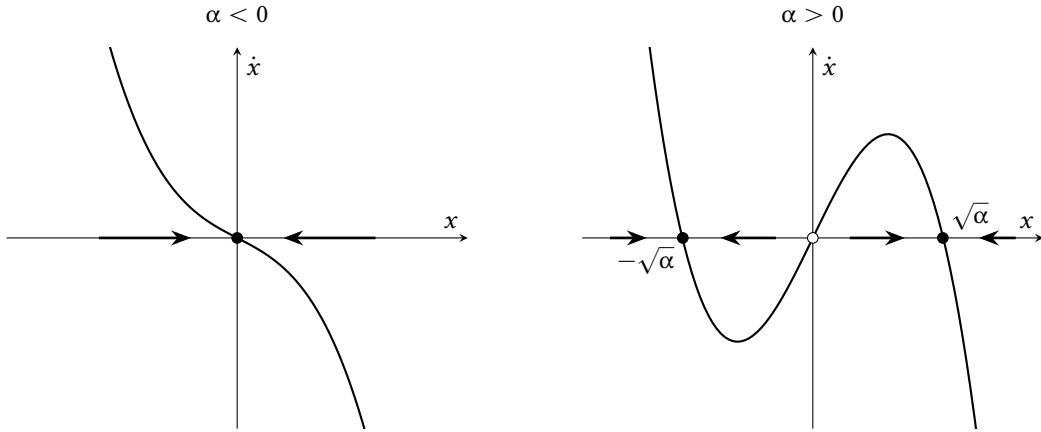


FIGURE 2. Plot of the phase space flow for the pitchfork bifurcation. The slope at the equilibrium points determines their stability: a negative slope indicates a sink (stable), while a positive slope indicates a source (unstable).

Such behaviour arises for instance in the study of particles systems: for example, in the Curie-Weiss model, the average magnetisation of the system satisfies:

$$\frac{dm}{dt} = -m + \tanh \beta m,$$

it in fact undergoes a pitchfork bifurcation at a critical value of the inverse temperature β and can be seen as a gradient flow of the Helmholtz free energy

$$f(m) \approx \frac{1}{2\beta} m^2 + \frac{1}{12\beta} m^4$$

thus we can identify the bifurcation with a phase transition occurring in the system as the temperature of the system is varied. [KRB10, Sect. 8.3]

With a microscopic point of view, one can instead look at trajectories of an individual particle, governed by Newton's equation, now with an additional linear dissipation term, driven by some friction coefficient γ as before, and a stochastic forcing driven by the Brownian noise $\xi_t = \sigma \dot{W}_t \equiv \sqrt{2\gamma\beta^{-1}} \dot{W}_t$:

$$\ddot{x}_t = -\nabla V(x_t) - \gamma \dot{x}_t + \xi_t.$$

This is the Langevin equation, governing the diffusive behaviour of a particle in the presence of a thermal reservoir at temperature β . [Pav14, Sect. 6.1] Again, we can consider the overdamped limit:

$$dx_t = -\frac{1}{\gamma} \nabla V(x_t) dt + \frac{\sigma}{\gamma} dW_t,$$

recovering exactly the deterministic gradient flow equation as $\sigma \downarrow 0$. Going back to our original one-dimensional example, we consider the class of Stochastic Differential Equations

$$dx_t = (\alpha x_t - x_t^3) dt + \sigma dW_t \quad \alpha \in \mathbb{R}$$

where $(W_t)_{t \in \mathbb{R}}$ denotes the standard Wiener process on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The natural question then is, whether the same dynamical behaviour can be observed for the system when such additive noise is added, i.e. whether this additive noise preserves the pitchfork bifurcation or not. By providing a formulation of the system as a Random Dynamical System and relying on their theory of attractors for such systems, Hans Crauel and Franco Flandoli argued that the latter is the case, i.e. the bifurcation is destroyed. [CF98]

On the other hand, this view was later challenged by Jeroen Lamb, Martin Rasmussen et al. [Lam+17] who argued that a qualitative change in the dynamics is still present, by observing the dichotomy spectrum associated to the system.

We shall first study precisely the initial system without noise, by recalling the classic theory of Deterministic Dynamical Systems and their bifurcations, so that the precise mathematical setting, i.e. the nature of the space X considered (even though in our toy example this will just be $\mathbb{R}$) is specified.

Then, we shall introduce the general theory of Random Dynamical Systems, which was developed in the 1990s, loosely with the perspective of the joint dynamics on the space X and the probability space Ω . We refer to the unified treatment of Ludwig Arnold [Arn98] and to Crauel and Flandoli for the crucial notion of random attractors [CF94], which will play a fundamental role in their argument, which we then present.

Finally, we shift our perspective to the dichotomy spectrum, by looking at the original approach by Robert Sacker and George Sell [SS78] and the arguments against destruction of the bifurcation in our example.

We shall study some notions in a greater generality than required, as similar questions are still studied in the research today in a variety of settings, for instance regarding pitchfork bifurcations for Stochastic Partial Differential Equations (e.g. [BEN23; BPZ26]) in which case, the interpretation of the bifurcation as a phase transition of the underlying physical model has profound consequences.

1. DETERMINISTIC DYNAMICAL SYSTEMS AND THEIR PITCHFORK BIFURCATIONS

We study the one-dimensional gradient flow

$$\dot{x}_t = -V'(x_t), \quad V(x) = -\frac{\alpha}{2}x^2 + \frac{1}{4}x^4, \quad \alpha \in \mathbb{R}$$

which we can view as an autonomous differential equation on $X = \mathbb{R}$. Mathematically, we would like to develop a framework to study such a system with an arbitrary degree of generality, up to the case in which X is no more than a topological space. Here we want to show that the setting in which X is a smooth manifold is particularly convenient, mainly because the choice of coordinates is non-ambiguous and one has a clear formulation for the dynamical systems and its behaviour. This choice will turn out to be even more convenient for the description of the dichotomy spectrum later.

1.1. Why manifolds are the right place to study dynamical systems. Let $X = \mathbb{R}^2$ and consider the gradient flow with potential

$$V(x, y) = -\frac{\alpha}{2}(x^2 + y^2) + \frac{1}{4}x^4, \quad \alpha \in \mathbb{R}$$

yielding the pair of ODEs:

$$\dot{x}_t = -\alpha x_t + x_t^3, \quad \dot{y}_t = -\alpha y_t.$$

Now, suppose we perform a coordinate change to polar coordinates, i.e. $(x, y) \mapsto (r \cos \phi, r \sin \phi)$, so that our potential is given by

$$V(r, \phi) = -\frac{\alpha}{2}r^2 + \frac{1}{4}r^4 \cos^4 \phi, \quad \alpha \in \mathbb{R}.$$

Then, the form of the equations and thus of the trajectories it generates in the phase space is not preserved if we take the gradient flow as it is, and we need to ‘correct’ the gradient to recover those. [Mie23]

The correction term is given by the Jacobian of the change of coordinates, in fact

$$\begin{pmatrix} 1 & 0 \\ 0 & 1/r^2 \end{pmatrix} \nabla V(r, \phi) = \nabla V(x, y) \Big|_{x=r \cos \phi, y=r \sin \phi}$$

One geometric perspective is to view the Euclidean gradient as preceded by the identity matrix, representing the flat metric on $\mathbb{R}^2$, and similarly consider the correction term for the polar gradient as the inverse metric as changed by the coordinate transformation. This is the usual way of taking the gradient in Riemannian geometry: given a real-valued map V over a manifold X , its gradient is given by $dV^\#$, that is the musical isomorphism induced by the metric.

1.1.1. (Semi)-flows on manifolds. The geometric formalism is also convenient because in autonomous equations we are just imposing the velocity $\dot{x}_t$ of a curve equal to a vector field $\mathcal{V} : X \rightarrow X$, given by the gradient of the

potential in this context, and we have a very natural geometric description of vector fields, which we now recall. We refer to the classical textbook by John Lee [Lee03].

Consider the tangent bundle $TX = \bigsqcup_{p \in X} T_p X$ i.e., the collection of all tangent vectors at every point on the manifold, and *sections* of it, $\mathcal{V} : X \rightarrow TX$, associating to each point: $p \mapsto \mathcal{V}_p \in T_p X$ a tangent vector. This is a *vector field*, and we denote by $\mathfrak{X}(X)$ the collection of them. We say that $\varphi : [0, \infty) \rightarrow X$ is an *integral curve* of the vector field $\mathcal{V} \in \mathfrak{X}(X)$ if:

$$\dot{\varphi}_t = \mathcal{V}_{\varphi_t} \text{ for all } t \geq 0 \quad (\text{resp. } t \in \mathbb{R})$$

that is, at each point p , its velocity is exactly $\mathcal{V}_p$. Now, by the existence theorem for ODEs on the coordinate representation of the smooth vector field $\mathcal{V}$, for each point $p \in X$, there exists an integral curve $\varphi_p : I \subset \mathbb{R} \rightarrow X$ starting at p , i.e. $\varphi_p(0) = p$. For each $t \geq 0$, this gives a endomorphism $\varphi_t : p \in X \mapsto \varphi_t(p) \equiv \varphi_p(t) \in X$ ‘sliding’ the manifold along integral curves. Since, by the chain rule, $\varphi_p(t+s)$ is an integral curve at $q \equiv \varphi_p(s)$, assuming also uniqueness of integral curves, $\varphi_q(t) = \varphi_p(t+s)$ and thus $\varphi_t \circ \varphi_s(p) = \varphi_{t+s}(p)$, meaning that we can define a *global (semi)-flow on X* as the continuous map $\varphi : [0, \infty) \times X \rightarrow X$ such that $\varphi_t(\varphi_s(p)) = \varphi_{t+s}(p)$ and $\varphi_0(p) = p$. The collection of homeomorphisms $\varphi_t \equiv \varphi_t(p)$ satisfies the *semigroup property*

$$\varphi_t \circ \varphi_s = \varphi_{t+s} \quad \varphi_0 = \text{id}_X$$

as the image of the curve $\varphi_p(t) \equiv \varphi_t(p)$ is the orbit of p under the group action φ of the additive group $\mathbb{R}_+$ on X .

1.2. Definition of a Deterministic Dynamical System. The previous discussion motivates the following definition, presented in analogy to [Arn98; CF94].

DEFINITION 1.1. Suppose $T = \mathbb{R}, \mathbb{R}_+, \mathbb{Z}$ or $\mathbb{N}$, and let (X, dist) be a metric space. A (smooth) *deterministic dynamical system* with time T on X is a map

$$\varphi : T \times X \rightarrow X, \quad (t, x) \mapsto \varphi_t x$$

satisfying:

- i. continuity: $x \mapsto \varphi_t x$ is continuous (resp. differentiable) for every $t \in T$;
- ii. identity: $\varphi_0 x = x$ for all $x \in X$, meaning $\varphi_0 = \text{id}_X$;
- iii. (semi)-group property: for all $t, s \in T$ and $x \in X$,

$$\varphi_{t+s} x = \varphi_t \circ \varphi_s x.$$

Given any $x \in X$, the *orbit*, or *trajectory*, of φ starting at x is the set $\bigcup_{t \in T} \varphi_t x$. [Tem12]

This definitely is far too general and we restrict to the case in which X is a smooth manifold and $T = [0, \infty)$, so that orbits are smooth submanifolds of X .

1.2.1. Deterministic attractors. We speak of *equilibrium* for a deterministic dynamical system φ for fixed points of the semigroup, i.e. $x \in X$ such that

$$\varphi_t x = x, \quad \forall t \geq 0.$$

DEFINITION 1.2 ([TEM12]). Let x^* be an equilibrium for φ , then its

- *stable manifold* $\mathcal{M}_-(x)$ is the (possibly empty) set of points $x \in X$ which belong to an orbit $\{x_t\}_{t \geq 0}$ and that, as initial conditions $x = x_{t_0}$, ‘eventually flow into the equilibrium’, i.e.

$$\varphi_{t-t_0}x \longrightarrow x^* \quad \text{as } t \longrightarrow \infty;$$

- *unstable manifold* $\mathcal{M}_+(x)$ is the (possibly empty) set of points $x \in X$ which belong to an orbit $\{x_t\}_{t \geq 0}$ and such that

$$x_t \longrightarrow x^* \quad \text{as } t \longrightarrow 0.$$

An equilibrium x^* is a *stable equilibrium* if $\mathcal{M}_+(x^*) = \emptyset$ and an *unstable equilibrium* otherwise. Finally, an *attractor* is a set $\mathcal{A} \subset X$ such that:

- $\mathcal{A}$ is an *invariant set*, i.e. $\varphi_t \mathcal{A} = \mathcal{A}$ for all $t \geq 0$;
- $\mathcal{A}$ possesses an open neighborhood $\mathcal{U}$, which is called *its basin of attraction*, such that, for every $x \in \mathcal{U}$:

$$\text{dist}(\varphi_t x, \mathcal{A}) \longrightarrow 0 \quad \text{as } t \longrightarrow \infty.$$

In particular, we speak of a *global attractor* when $\mathcal{A}$ is compact and its basin of attraction is the whole X .

1.3. Pitchfork bifurcation in the deterministic system. We now precisely detect the bifurcation for the class of ODEs

$$\dot{x}_t = -V'(x_t), \quad V(x) = -\frac{\alpha}{2}x^2 + \frac{1}{4}x^4, \quad \alpha \in \mathbb{R}$$

relying on the machinery developed above. As α crosses the critical value of 0, because $\mathcal{M}_+(0) = \emptyset$ (as it is a minimum, and there can be no orbit starting at this point), the dynamical regime changes from presenting a unique stable equilibrium, which is thus also a global singleton attractor, to having a couple of new stable equilibria created where the stability of the origin changes, in fact $\mathcal{M}_+(0) = (-\sqrt{\alpha}, \sqrt{\alpha})$ for all $\alpha > 0$.

LEMMA 1.3. $[-\sqrt{\alpha}, \sqrt{\alpha}]$ is an invariant set of φ whenever $\alpha > 0$.

Proof. Let $\alpha > 0$. We want to show $\varphi_t[-\sqrt{\alpha}, \sqrt{\alpha}] = [-\sqrt{\alpha}, \sqrt{\alpha}]$ for all $t \geq 0$.

The interval $[-\sqrt{\alpha}, \sqrt{\alpha}]$ contains three equilibria: $-\sqrt{\alpha}$, 0, and $\sqrt{\alpha}$.

- For $x \in (0, \sqrt{\alpha})$, we have $x > 0$ and $(\alpha - x^2) > 0$ and thus $\dot{x} > 0$. The flow pushes points to the right, towards $\sqrt{\alpha}$, but cannot cross the equilibrium $\sqrt{\alpha}$.
- For $x \in (-\sqrt{\alpha}, 0)$, we have $x < 0$ and $(\alpha - x^2) > 0$ and thus $\dot{x} < 0$. The flow pushes points to the left, towards $-\sqrt{\alpha}$, but cannot cross the equilibrium $-\sqrt{\alpha}$.

Because the boundary points are equilibria, no trajectory can exit the interval. Furthermore, the unstable manifold of the origin, $\mathcal{M}_+(0) = (-\sqrt{\alpha}, \sqrt{\alpha})$, ensures that the interior of the interval is ‘filled’ by the forward flow of points arbitrarily close to zero. Thus, the set maps exactly onto itself under the semi-flow. $\square$

LEMMA 1.4. The basin of attraction of the set $\mathcal{A}_\alpha = [-\sqrt{\alpha}, \sqrt{\alpha}]$ is the whole real line.

Proof. We must show that for any $x \in \mathbb{R}$, $\text{dist}(\varphi_t x, \mathcal{A}_\alpha) \longrightarrow 0$ as $t \longrightarrow \infty$.

- If $x \in [-\sqrt{\alpha}, \sqrt{\alpha}]$, the point is already in $\mathcal{A}_\alpha$, so the distance is null for all $t \geq 0$.
- If $x > \sqrt{\alpha}$, then $x^2 > \alpha$, meaning $(\alpha - x^2) < 0$. Since $x > 0$, the velocity $\dot{x} < 0$. The trajectory monotonically decreases toward the equilibrium $\sqrt{\alpha}$.
- If $x < -\sqrt{\alpha}$, then $x^2 > \alpha$, meaning $(\alpha - x^2) < 0$. Since $x < 0$, the velocity $\dot{x} > 0$. The trajectory monotonically increases toward the equilibrium $-\sqrt{\alpha}$.

For any point outside the interval, the flow drives it asymptotically toward the boundary of the interval. Therefore, $\lim_{t \rightarrow \infty} \text{dist}(\varphi_t x, \mathcal{A}_\alpha) = 0$ for all $x \in \mathbb{R}$. $\square$

THEOREM 1.5 ([CF98]). For each fixed $\alpha \in \mathbb{R}$, the system $\dot{x}_t = \alpha x_t - x_t^3$ has a global attractor, and the family of attractors $\{\mathcal{A}_\alpha\}_{\alpha \in \mathbb{R}}$ is given by

$$\mathcal{A}_\alpha = \begin{cases} \{0\} & \text{for } \alpha \leq 0 \\ [-\sqrt{\alpha}, \sqrt{\alpha}] & \text{for } \alpha > 0. \end{cases}$$

Proof. We first consider the case $\alpha \leq 0$.

The singleton $\{0\}$ is compact and, since $\dot{x}_0 = 0(\alpha - 0^2) = 0$, the point $x^* = 0$ is an equilibrium and thus $\{0\}$ is an invariant set. For any $x \neq 0$, evaluate the sign of the velocity $\dot{x} = x(\alpha - x^2)$. Because $\alpha \leq 0$ and $x^2 > 0$, the term $(\alpha - x^2)$ is strictly negative: if $x > 0$, then $\dot{x} < 0$ and the trajectory strictly decreases toward the origin; if instead $x < 0$, then $\dot{x} > 0$ and the trajectory strictly increases toward 0. Since the flow directs all initial conditions monotonically toward the origin, $\varphi_t x \rightarrow 0$ as $t \rightarrow \infty$ for all $x \in \mathbb{R}$. The basin of attraction is thus the whole real line, proving that $\{0\}$ is the global attractor.

Now, let $\alpha > 0$. Since, $[-\sqrt{\alpha}, \sqrt{\alpha}]$ is compact, we get the claim because of Lemma 1.3 and Lemma 1.4. $\square$

Notice that all the previous results never specify the analytic form of the semigroup generated by the system. Namely, there is no need to actually solve the autonomous ODE and the whole section can be viewed as a rigorous version of the qualitative analysis corresponding to Figure 2.

2. ADDITIVE NOISE DESTROYS A PITCHFORK BIFURCATION

Here, we want to argue that, in the words of Crauel and Flandoli, “as soon as additive white noise of arbitrarily small intensity is incorporated in the model, the bifurcation disappears and the dynamical behaviour of the disturbed system remains the same for all parameter values”. [CF98]

More precisely, we study the class of Stochastic Differential Equations

$$dx_t = -V'(x_t) dt + \sigma dW_t \quad V(x) = -\frac{\alpha}{2}x^2 + \frac{1}{4}x^4, \quad \alpha \in \mathbb{R}, \quad \sigma \geq 0 \quad (\text{SDE})$$

for trajectories $t \in T = [0, \infty) \mapsto x_t \in X = \mathbb{R}$, where each $x_t(\omega)$ is a random variable for noise realisations $\omega \in \Omega$, representing an entire, infinite trajectory of noise over all time. We thus endow Ω with the σ -algebra $\mathcal{F}$, allowing us to measure sets of these trajectories, and a probability measure $\mathbb{P}$ telling us how likely a certain set of trajectories is to occur in reality. Then, $(W_t)_{t \in \mathbb{R}}$ denotes the standard Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$. We model the evolution of noise via a family of measure-preserving transformations (so that the noise is, in a sense, ‘stationary’ over time) $\vartheta_t : \Omega \rightarrow \Omega$ such that the mapping $(t, \omega) \in T \times \Omega \mapsto \vartheta_t \omega \in \Omega$ is $\mathcal{B}(T) \otimes \mathcal{F}$ measurable,

where $\mathcal{B}$ denotes the Borel σ -field. It is then convenient to assume that ϑ itself constitutes a (deterministic) dynamical system, so that a variety of notions applies for this model of noise.

The dynamical system φ generated by (SDE) is however not autonomous any more, because it depends on the shifting noise. We recover an autonomous system by looking at the dynamical system Θ acting on the fibre bundle* $\Omega \times X$, with base space the measurable space Ω and, as fibres, copies of the manifold X attached at every realisation $\omega \in \Omega$. We namely study the so-called *skew-product* flow

$$\Theta_t(\omega, x) = (\vartheta_t \omega, \varphi_t(\omega)x)$$

consisting of a base dynamics ϑ_t that moves along the base space Ω , representing the autonomous evolution of the noise, and a fibre dynamics $\varphi_t(\omega)$ moving the state within the fibres, i.e. the copies of the manifold. The latter depends on the base point ω , rendering the overall dependency ‘skewed’. This skew-product is a completely autonomous semigroup over the bundle:

$$\Theta_{t+s}(\omega, \cdot) = \Theta_t \circ \Theta_s(\omega, \cdot)$$

$$(\vartheta_{t+s}\omega, \varphi_{t+s}(\omega)x) = (\vartheta_t(\vartheta_s\omega), \varphi_t(\vartheta_s\omega) \circ \varphi_s(\omega)x)$$

and we say that φ satisfies the *cocycle property*, this justifying the assumption that ϑ is itself a semigroup.

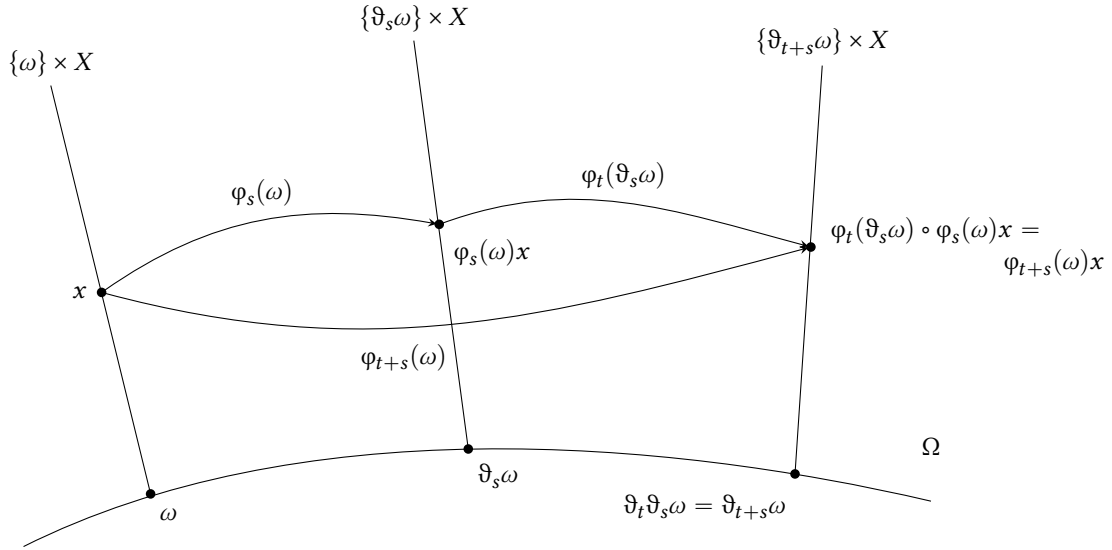


FIGURE 3. A random dynamical system as an action on a bundle. Image reproduced from [Arn98, Fig. 1.2].

2.1. Definition of a Random Dynamical System. The previous discussion motivates the following definition, presented as in [Arn98; CF94].

*See the appendix for the definition of fibre bundle.

DEFINITION 2.1. Suppose $T = \mathbb{R}, \mathbb{R}_+, \mathbb{Z}$ or $\mathbb{N}$, and let (X, dist) be a Polish space. A (smooth) *random dynamical system*, with time T on X over a dynamical system ϑ on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$, is a $\mathcal{B}(T) \otimes \mathcal{B}(X) \otimes \mathcal{F}$ measurable map

$$\varphi : T \times X \times \Omega \longrightarrow X, \quad (t, x, \omega) \longmapsto \varphi_t(\omega)x$$

satisfying:

- i. continuity: $x \mapsto \varphi_t(\omega)x$ is continuous (resp. differentiable) for every $t \in T$;
- ii. identity: $\varphi_0(\omega)x = x$ for all $x \in X$, meaning $\varphi_0(\omega) = \text{id}_X$;
- iii. cocycle property: for all $t, s \in T$ and $\omega \in \Omega$,

$$\varphi_{t+s}(\omega) \equiv \varphi_t(\vartheta_s \omega) \circ \varphi_s(\omega).$$

Again, we restrict to the case in which X is a smooth manifold and $T = \mathbb{R}$.

2.1.1. *Random attractors.* We consider set-valued maps $\mathcal{A} : \Omega \longrightarrow \text{closed}(X)$, which are said to be *measurable* if, for all $x \in X$, $\omega \mapsto \text{dist}(x, \mathcal{A}(\omega))$ is a measurable function. Such a measurable map $\mathcal{A}$ can be considered as a *closed random set*.

For the following definition, consider a random dynamical systems (φ, ϑ) with two-sided time $T = \mathbb{R}$ or $\mathbb{Z}$.

DEFINITION 2.2 ([CF94]). A closed random set $\mathcal{A}$ is a *global attractor* for (φ, ϑ) if

- i. $\mathcal{A}$ is compact for $\mathbb{P}$ -a.e. $\omega \in \Omega$;
- ii. $\mathcal{A}$ is (*forward*) *invariant*, i.e. for all $t > 0$ and $\mathbb{P}$ -a.e. $\omega \in \Omega$:

$$\varphi_t(\omega)\mathcal{A}(\omega) = \mathcal{A}(\vartheta_t \omega);$$

- iii. $\mathcal{A}$ *attracts* every bounded deterministic set $B \subset X$, i.e. the system exhibits *pullback attraction*

$$\text{dist}(\varphi_t(\vartheta_{-t} \omega)B, \mathcal{A}(\omega)) \longrightarrow 0 \quad \mathbb{P} - \text{a.s.}, \quad \text{as } t \longrightarrow \infty.$$

Both the random equivalent of the invariance and attracting notion must now account for the base noise dynamics given by ϑ : in particular, being an invariant set corresponds to being pushed forward by the fibre dynamics φ_t exactly to the corresponding state of the base dynamics and, similarly, to detect an attractor we pull back into the past the base dynamics ϑ_{-t} and then follow the long-time behaviour of fibre dynamics from there.

Here, as before, we omit the discussion of existence and uniqueness of a (connected) global attractor.

2.2. Random Dynamical Systems induced by Stochastic Differential Equations. Let us now look at the random dynamical system induced by (SDE). [CF98]

By local Lipschitz continuity, it has a unique solution for any initial condition. However, solutions can only exist for positive times: indeed, even in the deterministic case $\sigma = 0$, $\dot{x}_t = -V'(x)$ blows up in finite time. Let us prove that this is the case.

Recall that, given any stochastic differential equation $dx_t = \mu(x_t) dt + \sigma dW_t$, there is a corresponding diffusion process with infinitesimal generator given by the elliptic differential operator

$$\mathcal{L}f(x) = \mu(x)\nabla f(x) + \frac{\sigma^2}{2}\Delta f(x).$$

LEMMA 2.3. For any finite time $t > 0$, the (unique) solution to (SDE) exists.

Proof. We invoke [Kha11, Thm. 3.5], ensuring that, if there exists a smooth (Lyapunov) function f such that, for some constant $c > 0$:

- (1) $\mathcal{L}f \leq cf$;
- (2) $\inf_{|x|>r} f(x) \rightarrow \infty$ as $r \rightarrow \infty$;

then there exists an a.s. continuous solution to the stochastic differential equation associated to $\mathcal{L}$. We choose $f(x) = V(x)$ so that condition (2) is trivially satisfied, as $\lim_{|x| \rightarrow \infty} f(x) = \infty$, and, since $\mu(x) = -V'(x)$ in (SDE),

$$\mathcal{L}V(x) = -V'(x)^2 + \frac{\sigma^2}{2}V''(x) = -(-\alpha x + x^3)^2 - \frac{\sigma^2}{2}(\alpha - 3x^2) = -x^6 + 2\alpha x^4 - \left(\alpha^2 - \frac{3\sigma^2}{2}\right)x^2 - \frac{\alpha\sigma^2}{2}.$$

As a consequence, $\lim_{|x| \rightarrow \infty} \mathcal{L}V(x) = -\infty$ and thus, by continuity, $\mathcal{L}V$ must be bounded from above. $\square$

Whence, the random dynamical system entailed by (SDE) must have the real line as state space and, more problematically, one-sided time $T = [0, \infty)$. This is not convenient because Def. 2.2 does not directly apply.

Luckily, there is a result, due to Crauel, establishing a one-to-one correspondence between invariant measures for φ with respect to a two-sided time base flow and invariant measures with respect to a restriction of ϑ_t to one-sided time. [Cra90, Sect. 3]

2.2.1. *Invariant measures of Random Dynamical Systems.* We reduce to the well-established concept of invariant measure for a deterministic dynamical system.

DEFINITION 2.4 ([CRA90]). A measure $\rho \in \mathcal{P}(\Omega \times X)$, with marginal $\mathbb{P}$ on Ω , is said to be an *invariant measure* for the random dynamical system (φ, ϑ) if it is invariant under the associated skew product flow Θ , i.e.

$$\Theta_t \rho = \rho, \quad \text{for all } t \in T.$$

We are again not interested with the problem of existence for such objects. We however want to point out a nice result [Cra90, Lemma 2.3], establishing existence of invariant measures for a random dynamical systems on a compact manifold, as a direct application of the Riesz-Markov-Kakutani representation theorem.

Consider the random measure $\omega \in \Omega \mapsto \rho(\omega) \in \mathcal{P}(\Omega \times X)$, and the following fibre-wise condition:

$$\mathbb{E}[\varphi_t \rho \mid \vartheta_t^{-1} \mathcal{F}](\omega) = \rho(\vartheta_t \omega), \quad \mathbb{P} - \text{a.s.}$$

where the conditional expectation is taken with respect to the σ -algebra $\vartheta_t^{-1} \mathcal{F} = \{\vartheta_t^{-1} B \mid B \in \mathcal{F}\}$.

Clearly, if the base dynamics were invertible, this condition would be equivalent to forward invariance for the stationary measure:

$$\varphi_t \rho(\omega) = \rho(\vartheta_t \omega).$$

However, noise is often non-invertible, as multiple different past states could lead to the exact same future outcome. Intuitively, because we do not know which past led to the current state, we must ‘average out’ the possibilities. Now, given the invariant measure ρ for the two-sided (φ, ϑ) , consider the σ -field $\mathcal{F}^+ = \sigma\{\varphi_t \mid t > 0\}$, the smallest one that makes $\omega \mapsto \varphi_t(\omega)$ measurable and such that,

$$\vartheta_t^{-1}\mathcal{F}^+ = \sigma\{\varphi_t \circ \vartheta_s \mid t > 0\} = \sigma\{\varphi_{t+s} \circ \varphi_s^{-1} \mid t > 0\} \subset \mathcal{F}^+$$

for any $s \geq 0$, by the cocycle property. For instance, the natural Brownian field $\mathcal{F}^+ = \sigma\{W_t \mid t > 0\}$ is the natural choice for (SDE). Then, we can restrict (φ, ϑ) to a one-sided random dynamical system with $(\Omega, \mathcal{F}^+, \mathbb{P}|_{\mathcal{F}^+})$, with invariant measure given by $\rho^+ = \rho|_{\mathcal{F}^+ \otimes \mathcal{B}}$.

If we were able to invert this restriction, we would have a full correspondence between ρ and ρ^+ .

LEMMA 2.5. Consider the operator over $\mathcal{P}(\Omega \times X)$ given by

$$\bar{\Theta}_t \rho = \varphi_t^{-1}(\rho \circ \vartheta_t).$$

Then, $\bar{\Theta}_t$ is a flow and a measure ρ is invariant for the two-sided (φ, ϑ) if and only if $\bar{\Theta}_t \rho = \rho$ for all $t \in T$.

Proof. First, we show the semigroup property: $\bar{\Theta}_s \circ \bar{\Theta}_t = \bar{\Theta}_{t+s}$. For any $s, t \geq 0$ and any random measure ρ ,

$$\bar{\Theta}_s(\bar{\Theta}_t \rho)(\omega) = \varphi_s^{-1}(\omega) \bar{\Theta}_t \rho(\vartheta_s \omega) = \varphi_s^{-1}(\omega) \varphi_t^{-1}(\vartheta_s \omega) \mu(\rho_t(\vartheta_s \omega)) = \varphi_{t+s}^{-1}(\omega) \rho(\vartheta_{t+s} \omega) = \bar{\Theta}_{t+s} \rho(\omega),$$

where we have used that ϑ is a flow, and the cocycle property of the base dynamics: $\varphi_{t+s}^{-1}(\omega) = \varphi_s^{-1}(\omega) \circ \varphi_t^{-1}(\vartheta_s \omega)$. If ρ is invariant then, in particular $\varphi_t \rho(\omega) = \rho(\vartheta_t \omega)$ and thus

$$\bar{\Theta}_t \rho(\omega) = \varphi_t^{-1}(\omega) \rho(\vartheta_t \omega) = \varphi_t^{-1} \circ \varphi_t \rho(\omega) = \rho(\omega)$$

and conversely, if $\rho(\omega) = \bar{\Theta}_t \rho(\omega) = \varphi_t^{-1}(\omega) \rho(\vartheta_t \omega)$, this gives the desired claim. $\square$

LEMMA 2.6 ([CRA90]). Let ρ^+ be an invariant measure for a one-sided random dynamical system (φ, ϑ) . Then, $\rho = \lim_{t \rightarrow \infty} \bar{\Theta}_t \rho^+$ is an invariant measure for the corresponding two-sided random dynamical system.

Proof. Because φ_s acts as a continuous flow on the state space, the operator $\bar{\Theta}_s$ is continuous with respect to the weak topology on $\mathcal{P}(\Omega \times X)$ and thus for all $s \geq 0$:

$$\bar{\Theta}_s \rho = \bar{\Theta}_s \left(\lim_{t \rightarrow \infty} \bar{\Theta}_t \rho^+ \right) = \lim_{t \rightarrow \infty} \bar{\Theta}_s \left(\bar{\Theta}_t \rho^+ \right) = \lim_{t \rightarrow \infty} \bar{\Theta}_{t+s} \rho^+ = \rho.$$

by the preceding lemma, yielding that ρ is an invariant measure for the corresponding two-sided random dynamical system. $\square$

CORRESPONDENCE THEOREM. There is a one-to-one correspondence between the invariant measures for the one-sided random dynamical system and those invariant measures for the two-sided random dynamical system which are measurable with respect to $\sigma\{\vartheta_t^{-1}\mathcal{F}^+ \mid t > 0\}$.

COROLLARY 2.7 ([CF98]). Given a stochastic differential equation inducing a random dynamical system (φ, ϑ) , with one-sided time, and Markov semigroup $(P_t)_{t \geq 0}$, then there is a one-to-one correspondence between stationary measures for the associated Markov semigroup and invariant measures for the random dynamical system, given as follows:

- if ρ^+ is an invariant measure, let ρ be the invariant measure, for the two-sided random dynamical system, prescribed by the correspondence theorem. Then, the marginal of ρ on X , given by the disintegration

$$\mu = \int_{\Omega} \rho \, d\mathbb{P},$$

is a stationary measure;

- if μ is a stationary measure, then

$$\rho(\omega) = \lim_{t \rightarrow \infty} \varphi_t(\vartheta_{-t}\omega)\mu$$

is an invariant measure.

Fixing $\alpha \in \mathbb{R}$ and $\sigma \geq 0$, the stationary measure for the Markov semigroup induced by (SDE) is given by the canonical Boltzmann-Gibbs measure

$$d\mu(x) \propto e^{-2V(x)/\sigma^2} dx.$$

This density has, inherited from the potential, one maximum in 0 when $\alpha < 0$ and two maxima in $\pm\sqrt{\alpha}$ when $\alpha > 0$. We moreover have the convergences

$$\mu \longrightarrow \begin{cases} \delta_0 & \text{for } \alpha \leq 0 \\ (\delta_{-\sqrt{\alpha}} + \delta_{\sqrt{\alpha}})/2 & \text{for } \alpha > 0, \end{cases} \quad \text{as } \sigma \downarrow 0.$$

This peaked measure is the deterministic behaviour we have observed, and we already observe that, instead of fracturing into multiple isolated invariant measures, the stochastic system possesses exactly one unique invariant measure.

2.3. The arguments against a bifurcation. We now have a way to infer properties of invariance for a random dynamical system induced by a stochastic differential equation by looking at the associated Markov semigroup and its stationary measures. This is made rigorous by Cor. 2.10.

LEMMA 2.8 ([CF98]). Suppose (φ, ϑ) is a random dynamical system induced by a (forward complete, one-dimensional) stochastic differential equation such that the associated Markov semigroup has a unique stationary probability measure, and that $\mathcal{A}(\omega)$ is an invariant random compact set. Then, $\mathcal{A}$ is $\mathbb{P}$ -a.s. a singleton.

Proof. Consider the random variables $X^+(\omega) = \max \mathcal{A}(\omega)$ and $X^-(\omega) = \min \mathcal{A}(\omega)$ such that, by invariance:

$$\varphi_t X^+(\omega) = X^+(\vartheta_t \omega) \quad \text{and} \quad \varphi_t X^-(\omega) = X^-(\vartheta_t \omega), \quad \mathbb{P} - \text{a.s.}$$

for every $t > 0$. As a consequence, the random measures

$$\rho^+(\omega) = \delta_{X^+(\omega)} \quad \text{and} \quad \rho^-(\omega) = \delta_{X^-(\omega)}$$

are invariant. But then, by Cor. 2.7, both

$$\mu^+ = \int_{\Omega} \rho^+ d\mathbb{P} \quad \text{and} \quad \mu^- = \int_{\Omega} \rho^- d\mathbb{P}$$

are stationary measures for the Markov semigroup. However, we assumed that is unique, so $X^+(\omega) = X^-(\omega)$ for $\mathbb{P}$ -a.e. ω . $\square$

We can apply this theorem to get a result for (SDE) equivalent to Thm. 1.5, but with opposite conclusions.

THEOREM 2.9. For every $\alpha \in \mathbb{R}$ and $\sigma > 0$, the global attractor $\mathcal{A}_{\alpha}(\omega)$ is almost surely a singleton $\{a_{\alpha}(\omega)\}$.

By proving this theorem, Crauel and Flandoli effectively argue for the destruction of any bifurcating behaviour: in fact, as α varies and crosses the origin, we do not observe the global attractor ‘transforming’ from a singleton into the compact interval $[-\sqrt{\alpha}, \sqrt{\alpha}]$, but it rather remains a one-point set, and this is observed as soon as the noise $\sigma > 0$. If one were not yet convinced, we can show an even stronger collision, or *synchronisation*, property for orbits: for any $x_1, x_2 \in \mathbb{R}$,

$$|\varphi_t(\vartheta_{-t}\omega)x_1 - \varphi_t(\vartheta_{-t}\omega)x_2| \longrightarrow 0 \quad \text{as } t \longrightarrow \infty,$$

for $\mathbb{P}$ -a.e. $\omega \in \Omega$. In fact, for any bounded $B \in \mathcal{B}(\mathbb{R})$,

$$\lim_{t \rightarrow \infty} \text{dist}(\varphi_t(\vartheta_{-t}\omega)B, a(\omega)) = 0 \quad \mathbb{P} - \text{a.s.}$$

and by the triangle inequality:

$$\begin{aligned} |\varphi_t(\vartheta_{-t}\omega)x_1 - \varphi_t(\vartheta_{-t}\omega)x_2| &\leq |\varphi_t(\vartheta_{-t}\omega)x_1 - a(\omega)| + |a(\omega) - \varphi_t(\vartheta_{-t}\omega)x_2| \\ &= \text{dist}(\varphi_t(\vartheta_{-t}\omega)x_1, a(\omega)) + \text{dist}(\varphi_t(\vartheta_{-t}\omega)x_2, a(\omega)) \longrightarrow 0. \end{aligned}$$

This also the case in the deterministic system when $\alpha \leq 0$, in which context orbits follow geodesic paths towards the global minimum of the potential, but not for the double-well setting for $\alpha > 0$.

We can also quantify these properties in terms of the invariant measure.

COROLLARY 2.10. The random measure $\omega \mapsto \delta_{a(\omega)}$ is the unique invariant measure of (φ, ϑ) .

The claim follows from Crauel’s analogue to the Poincaré recurrence theorem for random dynamical systems [Cra99, Cor. 4.4], saying that, in the presence of a global attractor, one has concentration of the invariant measure on such random set.

3. ADDITIVE NOISE DOES NOT DESTROY A PITCHFORK BIFURCATION

We have seen that, in the deterministic system before the bifurcation, we have synchronisation and all trajectories inevitably collide at the minimum of the potential: the origin is a stable node. This is easily seen by linearising the gradient flow $\dot{x}_t = -V'(x_t)$, in fact $-V''(x^* = 0) < 0$. We are making implicit use of the Hartman-Grobman theorem to predict that the stability remains the same as for the linearised system. Someone, having doubts on Crauel and Flandoli’s arguments, may want to do the same for the random dynamical systems induced by (SDE).

They indeed allude [CF98, Rmk. 3.7] to the *linear random dynamical system*, in the sense that

$$\varphi_t(\omega)[ax + by] = a\varphi_t(\omega)x + b\varphi_t(\omega)y, \quad a, b \in \mathbb{R}, \quad t \in T, \quad \omega \in \Omega, \quad x, y \in X,$$

corresponding to the linearisation

$$\dot{v}_t = -V''(\varphi_t(\omega)x)v_t \tag{lin.}$$

of (φ, ϑ) along the orbit $\varphi_t(\omega)x$, and introduce the associated *Lyapunov exponent*

$$\begin{aligned} \lambda &= \lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t -V''(\varphi_\tau(\omega)x) d\tau, \quad \text{by the Itô formula:} \\ &= \lim_{t \rightarrow \infty} \left(-\frac{2}{\sigma^2 t} \int_0^t (V'(\varphi_\tau(\omega)x))^2 d\tau - \frac{2}{\sigma^2 t} (V(\varphi_t(\omega)x) - V(\varphi_0(\omega)x)) + \frac{2}{\sigma t} \int_0^t V'(\varphi_\tau(\omega)x) dW_\tau \right) < 0 \end{aligned}$$

concluding that any two solutions converge to each other *exponentially fast* (P-a.s.) for every value of the parameter $\alpha \in \mathbb{R}$.

However, this Lyapunov exponent is an asymptotic quantity and only describes what happens on average over a long time. It allows for the possibility that a particularly ‘rough’ noise path might cause the trajectories to diverge wildly for a while before finally coming together. A refined notion was introduced by [Lam+17, Sect. 4.2] and serves as the core of their first argument for a bifurcation.

DEFINITION 3.1. The global attractor $\{\varphi(\omega)\}_{\omega \in \Omega}$ is called *locally uniformly attractive* if there exists $\epsilon > 0$ such that

$$\lim_{t \rightarrow \infty} \sup_{x \in (-\epsilon, \epsilon)} \text{ess sup}_{\omega \in \Omega} |\varphi_t(\omega)[\varphi(\omega) + x] - \varphi(\vartheta_t \omega)| = 0.$$

If this is the case for some value of $\alpha \in \mathbb{R}$, then it is implied that attraction must happen irrespective of the noise realisation. However, Lamb and Rasmussen observed a qualitative change in the uniformity of attraction of $\{\varphi(\omega)\}_{\omega \in \Omega}$, suggesting the presence of a random bifurcation.

ASYMPTOTIC DESCRIPTION OF THE RANDOM PITCHFORK BIFURCATION. Let $\mathcal{A} = \{\varphi(\omega)\}_{\omega \in \Omega}$ be the random attractor of (SDE), then:

- for $\alpha < 0$, $\mathcal{A}$ is (globally) uniformly attractive with exponential rate:

$$|\varphi_t(\omega)x - \varphi_t(\omega)\varphi(\omega)| \leq e^{\alpha t} |x - \varphi(\omega)|, \quad \text{for all } x \in \mathbb{R};$$

- for $\alpha > 0$, $\mathcal{A}$ is not locally uniformly attractive.

The exponential attractivity estimate follows from Grönwall’s inequality applied to the integral version of (SDE). We can also, analogously to the (asymptotic) Lyapunov exponent considered by Crauel and Flandoli, propose the following *finite-time Lyapunov exponent*

$$\lambda_\tau(\omega) = \frac{1}{\tau} \log |D\varphi_\tau(\omega)|$$

where D denotes the differential operator over X and such that $\lambda = \lim_{\tau \rightarrow \infty} \lambda_\tau$.

FINITE-TIME DESCRIPTION OF THE RANDOM PITCHFORK BIFURCATION. For any finite time interval $[0, \tau]$, let λ_τ denote the finite-time Lyapunov exponent associated with (SDE), then:

- for $\alpha < 0$, the random attractor $\mathcal{A}$ is *finite-time attractive*, i.e.

$$\lambda_\tau \leq \alpha < 0, \quad \text{for all } \omega \in \Omega;$$

- for $\alpha > 0$, the random attractor $\mathcal{A}$ is not finite-time attractive, in fact

$$\mathbb{P}\{\omega \in \Omega : \lambda_\tau(\omega) > 0\} \neq 0.$$

This suggests that change in sign of the (finite-time) Lyapunov exponent marks a modification of attractivity and thus a (random) bifurcation point.

We will now look at the theory of Lyapunov exponents, which will also make precise how, by looking at the linearised system, we deduce something about a random dynamical system. The entire discussion will motivate the definition of dichotomy spectrum as a uniform version of the Lyapunov spectrum, and later provide a definitive rigorous argument for the presence of a random bifurcation.

3.1. Lyapunov exponents. Let ϑ denote a (deterministic) dynamical system over the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with time T that is:

- *measure-preserving*, i.e.

$$\vartheta_t^* \mathbb{P}\{A\} = \mathbb{P}\{\vartheta_t^{-1} A\} = \mathbb{P}\{A\}$$

for all $t \in T$ and $A \in \mathcal{F}$; and moreover

- *ergodic*, i.e. invariant sets have either full or null measure, namely for any $A \in \mathcal{F}$:

$$\vartheta_t^{-1} A = A \quad \text{for all } t \in T \quad \implies \quad \mathbb{P}\{A\} \in \{0, 1\}.$$

For such a system [Lam+17, App. A.1], given any *observable* $f \in L^1(\Omega)$, Birkhoff's pointwise ergodic theorem holds:

$$\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t f(\vartheta_\tau \omega) d\tau = \int_\Omega f d\mathbb{P}, \quad \text{for } \mathbb{P} - \text{a.e. } \omega.$$

In the discrete case in which $T = \mathbb{Z}$ or $\mathbb{N}$, the integral ‘space average’, on the left hand side, collapses an additive average and one can also consider multiplicative ones. [Fil19]

For instance, let $g = \exp(f)$, then for a.e. $\omega \in \Omega$ we have:

$$\lim_{n \rightarrow \infty} \log(g(\omega)g(\vartheta\omega) \cdots g(\vartheta^{n-1}\omega))^{1/n} = \int_\Omega f d\mathbb{P}.$$

This idea is useful for the study of random dynamical systems because, for instance, once we look at the linearisation (Φ, ϑ) of a random dynamical system (φ, ϑ) , where $\varphi_t(\omega) : X \rightarrow X$ and thus

$$\Phi_t(\omega)x = D\varphi_t(\omega)x : T_x X \rightarrow T_{\varphi_t(\omega)x} X$$

so that, by the chain rule, observing the system at discrete time steps yields the linear cocycle composition:

$$\Phi_n(\omega)x = \Phi_1(\vartheta^{n-1}\omega)\varphi_{n-1}(\omega)x \circ \dots \circ \Phi_1(\omega)x : T_x X \longrightarrow T_{\varphi_n(\omega)x} X$$

the asymptotic behaviour of which we are interested in.

Intuitively, we can imagine ‘dropping’ a perfectly round, infinitesimally small, sphere of initial conditions into our system. Mathematically, this sphere is the unit ball in the tangent space $T_x X$ at a starting point x . As the system evolves over n steps, the nonlinear dynamics φ_n carry the point x along its trajectory. Simultaneously, the linearised dynamics, i.e. the cocycle $\Phi_n(\omega)$, act on the tangent vectors. In fact, at each step, the linear cocycle $\Phi_1(\omega)$ takes the spherical ball of tangent vectors in the fibre over ω , and maps it to an initial ellipse in the tangent space attached to the new fibre over $\vartheta_1\omega$. Because Φ_n is still a linear operator, we can easily detect at how it ‘stretches, squashes, and rotates’ that perfect sphere into an ellipsoid of some kind, by looking at a decomposition of the bundle given by its eigenspaces.

This is what the multiplicative ergodic theorem of Oseledets essentially does, saying that as $n \longrightarrow \infty$, the lengths of the principal semi-axes of this ellipsoid grow or shrink exponentially, with rates of this exponential deformation given by a spectrum of *Lyapunov exponents* $\lambda_1 > \lambda_2 > \dots > \lambda_p$, for some $p \leq d$, where d denotes the dimension of the manifold X . The principal axes of this intuitive ellipsoid asymptotically align with specific subspaces in the tangent space, which we call *Oseledets spaces* $\mathcal{O}_1(\omega), \mathcal{O}_2(\omega), \dots, \mathcal{O}_p(\omega)$. For instance, if a ‘perturbation vector’ v lies in $\mathcal{O}_1$, it is caught in the ‘most expanding’ direction of the ellipse and, if it lies in the lowest $\mathcal{O}_p$, it is caught in the most contracting direction. Because the driving noise ω is ‘erratic’, the orientation of the Oseledets subspaces $\mathcal{O}_i(\omega)$ is entirely dependent on the specific noise realisation. The ellipse is constantly being rotated and re-oriented as it is whipped from fibre to fibre by the base flow ϑ .

Let X be a d -dimensional smooth Riemannian manifold, so that taking norms in the tangent bundle makes sense, as in the integrability condition below.

MULTIPLICATIVE ERGODIC THEOREM. [Arn98] Consider a random dynamical system (φ, ϑ) on X , where ϑ is an ergodic dynamical system on $(\Omega, \mathcal{F}, \mathbb{P})$ with time T , such that its linearisation $(\Phi = D\varphi, \vartheta)$ satisfies the integrability condition

$$\sup_{t \in [0,1]} \log^+ |\Phi_t|^{\pm 1} \in L^1(\Omega).$$

Then, there almost surely exist at most d Lyapunov exponents $\lambda_1 > \lambda_2 > \dots > \lambda_p$ and fibre-wise decomposition of the tangent bundle, called *Oseledets splitting* and given by the Whitney sum[†]

$$TX = \mathcal{O}_1(\omega) \oplus \mathcal{O}_2(\omega) \oplus \dots \oplus \mathcal{O}_p(\omega),$$

for Oseledets subspaces satisfying

$$\lim_{t \rightarrow \pm\infty} \frac{1}{|t|} \log |\Phi_t(\omega)x| = \lambda_i \quad \text{for all } x \in \mathcal{O}_i(\omega) \setminus \{0\}, \quad i \in \{1, \dots, p\}.$$

[†]See the appendix.

When the classic ergodic theorems, of Birkhoff or von Neumann, apply to a deterministic dynamical system, they intuitively tell us that all parts of the space in which the system moves will eventually be visited. In fact, ergodicity means exactly that all states are accessible: if there were invariant sets with neither full measure nor zero measure, then any trajectory starting in such a set would remain in it and the system would effectively be ‘walled off’ into two isolated regions. In an ergodic system this is not the case so a trajectory can never get trapped in a sub-region and, because the ergodic theorem establishes that time averages are equal to averaging over the state space, it is forced to eventually sample every part of the space proportionally to its measure.

With an ergodic base dynamics, the Oseledets theorem is instead prescribing, for a random dynamical system, a well-posed notion, i.e. the Lyapunov exponents, to determine ‘how aggressively’ (in terms of exponential divergence) the trajectory is being chaotic, or stable, while it is in the process of diffusing all over noise realisations.

3.1.1. The Lyapunov exponent of a one-dimensional stochastic differential equation. Now, we apply the multiplicative ergodic theorem to the system induced by (SDE), to finally justify in a rigorous way the asymptotic Lyapunov exponent λ introduced by Crauel and Flandoli.

For a random dynamical system (φ, ϑ) induced by a one-dimensional stochastic differential equation, the Oseledets theorem, on $X = \mathbb{R} \cong TX$, is equivalent to the ergodic theorem of Birkhoff. [Fil19, Rmk 2.2.11]

In fact, it is enough to consider the observable

$$f(\omega) = \log \frac{|\Phi_1(\omega)x|}{|x|}$$

so that the integrability condition of Oseledets implies $f \in L^1(\Omega)$. Moreover, the Oseledets splitting is trivial and we get a unique Lyapunov exponent. Intuitively, this is because we have one degree of freedom to expand or contract any spherical initial condition $[-\epsilon, \epsilon]$ into $[-e^{\lambda t}\epsilon, e^{\lambda t}\epsilon]$. For (SDE), the linearised system corresponds to (lin.), which has solution

$$v_t \propto \exp \left(\int_0^t -V''(\varphi_\tau(\omega)x) d\tau \right)$$

and thus we recover Crauel and Flandoli’s proposal for the Lyapunov exponent:

$$\begin{aligned} \lambda &= \lim_{t \rightarrow \infty} \frac{1}{t} \log |v_t| = \lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t -V''(\varphi_\tau(\omega)x) d\tau, & \text{by the ergodic theorem:} \\ &= - \int_{\Omega} \int_X V''(x) d\mu(x) d\mathbb{P}(\omega) < 0. \end{aligned}$$

3.2. The dichotomy spectrum for random dynamical systems. The multiplicative ergodic theorem establishes the decomposition

$$\Omega \times TX = \bigoplus_{i=1}^p \mathcal{O}_i$$

into subspaces that are invariant with respect to the skew-product flow Θ and are characterised dynamically as containing exactly those tangent vectors which have Lyapunov exponent λ_i . [Arn98]

All this machinery thus works perfectly when we can clearly split the behaviour of a system into two distinct directions: one where solutions shrink (decay) exponentially as time moves forward, and another where solutions shrink exponentially as time moves backward (which means they grow exponentially as time moves forward). That is, for instance, what happens in the presence of a hyperbolic equilibrium point (a saddle node) in autonomous systems and is the type of situation in which the Hartman-Grobman theorem applies: in fact, the real part of all eigenvalues of the linearised system $\Phi = D\varphi$ are nonzero, and the behaviour of the nonlinear system near the origin is topologically conjugate to the behaviour of the linear system. This is however totally lost for non-autonomous systems such as skew-product flows associated to random dynamical systems.

We first make rigorous the description of the well-behaved hyperbolic situation, by introducing the notion of systems possessing an exponential dichotomy, in the most general setting of fibre bundles, and then look at the dichotomy spectrum of Sacker and Sell, consisting of all those $\lambda \in \mathbb{R}$ for which $e^{-\lambda t}\Phi_t$ fails to have an exponential dichotomy, and which can thus be seen the non-autonomous equivalent of the real parts of eigenvalues.

DEFINITION 3.2 ([SS78]). A *projector* on the fibre bundle $\Omega \times X$ is defined to be a continuous mapping of each fibre into itself, on each of which it is a projection, i.e. $P^2(\omega) = P(\omega)$. Then, a linear random dynamical system (Φ, ϑ) is said to admit an *exponential dichotomy* with growth rate λ if there exists a projector P that is invariant, i.e. $P(\vartheta_t\omega)\Phi_t(\omega) = \Phi_t P(\omega)$ for all $t \in T$ and $\omega \in \Omega$, and satisfies:

$$\begin{aligned} |\Phi_t P(\omega)| &\leq K e^{(\lambda - \zeta)t}, & \text{for all } t \geq 0 \\ |\Phi_t [\text{id} - P](\omega)| &\leq K e^{(\lambda + \zeta)t}, & \text{for all } t \leq 0 \end{aligned}$$

for some constants $\zeta > 0$ and $K \geq 1$.

This is in analogy with the classical concept of dichotomy at an equilibrium point for an autonomous dynamical system $\dot{x} = Ax$, that is when A has no eigenvalues with a real part of zero, there is no ‘center’ behaviour and the entire state space X splits perfectly into two independent complementary subspaces and all trajectories are strictly forced exponentially ‘inwards’ or ‘outwards’.

If the multiplicative ergodic theorem applies, since the Lyapunov exponents λ_i represent the actual growth rates that exist in the system, none of them can equal the growth rate λ , and in fact this number must fall into the spectral gap:

$$\lambda \notin \{\lambda_1, \dots, \lambda_p\}.$$

We keep track of where λ falls among the Lyapunov exponents by $k = \max_i \{\lambda_i > \lambda\}$ and, because the null space of the dichotomy projector contains all the solutions that grow faster than the dichotomy rate, we have the splitting

$$\ker P(\omega) = \bigoplus_{i=1}^k \mathcal{O}_i(\omega)$$

for almost every $\omega \in \Omega$.

Consider the following subbundles of $\Omega \times X$,

$$\mathcal{S}_\lambda = \{(\omega, x) \in \Omega \times X : |e^{-\lambda t} \varphi_t(\omega)x| \rightarrow 0 \text{ as } t \rightarrow +\infty\}$$

$$\mathcal{U}_\lambda = \{(\omega, x) \in \Omega \times X : |e^{-\lambda t} \varphi_t(\omega)x| \rightarrow 0 \text{ as } t \rightarrow -\infty\}$$

respectively containing all initial states for which the flow grows strictly slower and faster than λ . [SS78]

In the presence of an exponential dichotomy, these two subbundles respectively correspond to the range and null set of the projector. [Lam+17, Prop. 2.5]

We can also provide a geometric intuition of the exponential dichotomy with rate λ , in the presence of which, $\mathcal{S}_\lambda$ and $\mathcal{U}_\lambda$ are *complementary invariant subbundles* of $\Omega \times X$, meaning that each fibre splits into:

$$\pi^{-1}(\omega) = \mathcal{S}_\lambda(\omega) \oplus \mathcal{U}_\lambda(\omega)$$

where π denotes the projection of the bundle[‡]. This is rigorously justified by the spectrum, that we now introduce.

DEFINITION 3.3 ([LAM+17]). The *dichotomy spectrum* Σ of the linear random dynamical system (Φ, ϑ) is the set of $\lambda \in \mathbb{R}$ for which (Φ, ϑ) does *not* admit an exponential dichotomy with growth rate $\lambda \in \mathbb{R}$.

By the discussion above about the Lyapunov spectral gap, the spectrum is a closed set containing neighbourhoods of each λ_i , for $i = 1, \dots, p$ and each so-called *spectral manifold*

$$\mathcal{W}_i = \mathcal{S}_{\lambda_i} \cap \mathcal{U}_{\lambda_{i-1}}$$

is an invariant subbundle. Each $\mathcal{W}_i$ in fact isolates a specific ‘band’ of growth and contains only the trajectories whose growth rates are strictly sandwiched between λ_{i-1} and λ_i . [SS78]

There is more: the dichotomy spectrum consists of at least one and at most d closed intervals. [Lam+17]

SPECTRAL THEOREM. Let (Φ, ϑ) be a linear random dynamical system with dichotomy spectrum Σ . Then, there exists $k \in \{1, \dots, d\}$ such that

$$\Sigma = [a_1, b_1] \cup \dots \cup [a_k, b_k], \quad \Omega \times TX = \bigoplus_{i=1}^k \mathcal{W}_i.$$

If the linear random dynamical system under consideration fulfills the conditions of the multiplicative ergodic theorem (that is, when the dichotomy spectrum is bounded), then the spectral manifolds of the above theorem are given by Whitney sums of Oseledets subspaces. Examples can be provided of linear random dynamical systems which satisfy the integrability condition of the multiplicative ergodic theorem, but which have no bounded dichotomy spectrum. [Lam+17, Ex. 3.8]

Unlike the Lyapunov spectrum, the dichotomy spectrum includes limits of the set

$$\lambda_\tau(\omega) = \frac{1}{\tau} \log |\Phi_t(\omega)|$$

of finite-time Lyapunov exponents (see [SS78, Thm. 3] and [Lam+17, Thm. 4.6]).

[‡]Refer to the appendix for the notions of projection and subbundles.

3.2.1. *The dichotomy spectrum at the bifurcation point.* In analogy with the classic situation, a random equilibrium is considered hyperbolic if and only if its dichotomy spectrum does not contain zero:

$$0 \notin \Sigma$$

In fact, a null growth rate represents the boundary between exponential decay and exponential growth and, if the spectrum does not contain the origin, a spectral gap exactly at the zero mark is guaranteed and there is a strict exponential dichotomy where every trajectory is forced to either strictly decay (stable) or strictly grow (unstable): as for deterministic dynamical systems, there are no neutral directions.

We now compute the dichotomy spectrum Σ_α of the linearisation around the unique attracting random equilibrium $\varphi_\alpha(\omega)$ of the system induced by (SDE), for each value of $\alpha \in \mathbb{R}$, observing that hyperbolicity is lost at the bifurcation point $\alpha = 0$. [Lam+17, Thm 4.5]

Let $\Phi_t^{(\alpha)}(\omega) = D\varphi_t(\omega)\varphi_\alpha(\omega)$ denote the linearised random dynamical system along $\varphi_\alpha(\omega)$.

THEOREM 3.4. The dichotomy spectrum Σ_α of $\Phi^{(\alpha)}$ is given by

$$\Sigma_\alpha = [-\infty, \alpha], \quad \text{for all } \alpha \in \mathbb{R}.$$

Proof. By integrating the linearised equation along $\varphi_\alpha(\omega)$:

$$\dot{\xi}(t) = (\alpha - 3\varphi_\alpha^2(\vartheta_t\omega)) \xi(t)$$

we get

$$\Phi_t^{(\alpha)}(\omega) = \exp \left(\int_0^t (\alpha - 3\varphi_\alpha^2(\vartheta_\tau\omega)) \, d\tau \right), \quad \left| \Phi_t^{(\alpha)}(\omega) \right| \leq e^{\alpha|t|} \quad \text{for all } t \in \mathbb{R} \quad (\star)$$

and thus $\Sigma_\alpha \subset [-\infty, \alpha]$.

We now argue by contradiction to show the converse implication: $[-\infty, \alpha] \subset \Sigma_\alpha$.

Suppose that $\Phi^{(\alpha)}$ admits an exponential dichotomy with growth rate $\lambda \in (-\infty, \alpha]$ and projector $P = \text{id}$ (the complementary case $P \equiv 0$ is treated analogously), i.e. $\Phi_t^{(\alpha)}(\omega) \leq Ke^{(\lambda-\zeta)t}$ for all values $t \geq 0$, and we fix one such value s.t. $e^{3t/4} > K$. But then, by attractivity [Lam+17, Prop. 4.1], there exists a positive measure set of noise realisations $\omega \in \Omega$ for which it holds:

$$\varphi_\alpha(\vartheta_s\omega) \in (-\sqrt{\zeta}/2, \sqrt{\zeta}/2) \quad \text{for all } s \in [0, t]$$

but this applied to $(\star)$ yields $\left| \Phi_t^{(\alpha)}(\omega) \right| \geq e^{t(\alpha-3\zeta/4)} > Ke^{(\lambda-\zeta)t}$, which is a contradiction. $\square$

APPENDIX

We here collect some facts about the theory of fibre bundles, as presented in the monograph by Husemöller on the subject. [Hus66]

Let Ξ , Ω and X be two connected topological spaces.

DEFINITION 3.5. Let Ξ , Ω and X be two connected topological spaces. Those three form a *fibre bundle*, with *base space* Ω and *fibre space* X and *total space* Ξ , if there exists a continuous and surjective *projection* map $\pi : \Xi \rightarrow \Omega$ that is *locally trivialising*, i.e. for any neighbourhood $\mathcal{O}$ there is a homeomorphism $f : \pi^{-1}(\mathcal{O}) \rightarrow \mathcal{O} \times X$ such that the following diagram commutes:

$$\begin{array}{ccc} \pi^{-1}(\mathcal{O}) & \xrightarrow{f} & \mathcal{O} \times X \\ \pi \downarrow & \swarrow \text{proj} & \\ \mathcal{O} & & \end{array}$$

where $\text{proj} : \mathcal{O} \times X \rightarrow \mathcal{O}$ is the natural projection. The preimage of each $\omega \in \Omega$, $\pi^{-1}(\omega)$ is thus homeomorphic to X and is called the *fibre* over ω .

If there exists a global such homeomorphism $f : \Xi \rightarrow \Omega \times X$, we speak of a *trivial bundle*.

A fibre bundle is a *subbundle* of Ξ if its total space and base space are, respectively, subspaces of Ξ and Ω , with projection given by the restriction of π on its total space.

All the bundles we have considered were trivial, but this is not always the case in the theory of random dynamical systems. For example, the Möbius band, which is not a product space and neither a trivial bundle, occurs as an invariant subbundle of $S^1 \times \mathbb{R}^2$ for some differential equation with periodic coefficients. [SS78]

DEFINITION 3.6 ([HUS66]). Consider two bundles Ξ_1 and Ξ_2 with the same base space Ω , then their *Whitney sum*, or *fibre product*, is the bundle

$$\Xi_1 \oplus \Xi_2 = \{(x, y) \in \Xi_1 \times \Xi_2 : \pi_1(x) = \pi_2(y)\}$$

with projection $\pi(x, y) = \pi_1(x) = \pi_2(y)$.

The name fibre product comes from the fact that all fibres of $\Xi_1 \oplus \Xi_2$ have the form $\pi_1^{-1}(\omega) \times \pi_2^{-1}(\omega) \subset \Xi_1 \times \Xi_2$. These notions are not only relevant to us because the description of a cocycle is rather natural when it is acting on the (trivial) fibre bundle $\Omega \times X$, but also and foremost because the multiplicative ergodic theorem establishes a fibre product decomposition of the fibre bundle $\Omega \times TX$, where we have an additional vector space structure on each fibre. This is an instance of the notion of *vector bundle*, of which the tangent bundle TX , of any smooth manifold X , is also an example. Notice that $\Omega \times X$ is not however a vector bundle, because X does not necessarily possess the structure of a linear space, though this was always the case in the particular application we have considered, as $X = \mathbb{R}$.

REFERENCES

- [Arn98] Ludwig Arnold. *Random Dynamical Systems*. Springer, 1998.
- [BEN23] Alex Blumenthal, Maximilian Engel, and Alexandra Neamțu. “On the pitchfork bifurcation for the Chafee–Infante equation with additive noise”. In: *Probability Theory and Related Fields* 187.3 (2023), pp. 603–627.
- [BPZ26] Alexandra Blessing, Nicolas Perkowski, and Chara Zhu. “Renormalization destroys a finite time bifurcation in the Φ_2^4 equation”. In: *arXiv:2602.08460* (2026).
- [CF94] Hans Crauel and Franco Flandoli. “Attractors for random dynamical systems”. In: *Probability theory and related fields* 100.3 (1994), pp. 365–393.
- [CF98] Hans Crauel and Franco Flandoli. “Additive noise destroys a pitchfork bifurcation”. In: *Journal of Dynamics and Differential Equations* 10.2 (1998), pp. 259–274.
- [Cra90] Hans Crauel. “Extremal Exponents of Random Dynamical Systems Do Not Vanish”. In: *Journal of Dynamics and Differential equations* 2.3 (1990), pp. 245–291.
- [Cra99] Hans Crauel. “Global random attractors are uniquely determined by attracting deterministic compact sets”. In: *Annali di Matematica pura ed applicata* 176.1 (1999), pp. 57–72.
- [Fil19] Simion Filip. “Notes on the multiplicative ergodic theorem”. In: *Ergodic Theory and Dynamical Systems* 39.5 (2019), pp. 1153–1189.
- [Hus66] Dale Husemöller. *Fibre bundles*. Springer, 1966.
- [Kha11] Rafail Khasminskii. *Stochastic Stability of Differential Equations*. Springer, 2011.
- [KRB10] Pavel L. Krapivsky, Sidney Redner, and Eli Ben-Naim. *A Kinetic View of Statistical Physics*. Cambridge University Press, 2010.
- [Lam+17] Jeroen S. W. Lamb et al. “The dichotomy spectrum for random dynamical systems and pitchfork bifurcations with additive noise”. In: *Annales de l’Institut Henri Poincaré, Probabilités et Statistiques* 53.4 (2017), pp. 1548–1574.
- [Lee03] John M. Lee. *Introduction to Smooth Manifolds*. Springer, 2003.
- [Mie23] Alexander Mielke. “An introduction to the analysis of gradients systems”. In: *arXiv:2306.05026* (2023).
- [Pav14] Grigorios A. Pavliotis. *Stochastic processes and applications*. Springer, 2014.
- [SS78] Robert J. Sacker and George R. Sell. “A spectral theory for linear differential systems”. In: *Journal of Differential Equations* 27.3 (1978), pp. 320–358.
- [Tem12] Roger Temam. *Infinite-dimensional dynamical systems in mechanics and physics*. Springer, 2012.